

Xiaoqi Wu

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Currently based in: London / Suzhou

Biopharma R&D | QC | Analytical | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Applied Molecular Biology, Genetics, Cell Biology and Genetics, Biochemistry and Microbiology, Bioinformatics, Programming and Statistics

Internship Experience

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Shanghai 2024.07 - 2024.09
Engineer · Analytical Department

- **Stability Testing:** Conducted stability testing of client-supplied drug samples using HPLC/UV, focusing on impurity detection and analysis
- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis, ensuring accuracy and reliability of test results
- **Quality Control:** Identified potential impurities through rigorous quality-control procedures, providing key data to support drug stability assessments – demonstrating solid technical skills in pharmaceutical analysis and instrument operation

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set, reduced from VecTraits' 91-column schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Combined PyMuPDF text extraction and Docling table-structure parsing to drive Gemma 4 31B IT (a 256K-context multimodal LLM) through 11 iterated prompt versions, reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Quality Verification:** Designed a bipartite record-matching evaluation framework and validated the locked pipeline on the held-out test set, achieving row precision of 1.000 and conditional macro field F1 of 0.729 – reflecting the same rigorous, systematic data-validation mindset applied in pharmaceutical quality control

Viral Fingerprinting: Population Genetics Analysis Using HERV-K solo-LTR Polymorphism London 2023.11 - 2023.12

PCR, gel electrophoresis, population genetics analysis

- **Lab Work:** Detected genomic polymorphisms across 5 primer sets using PCR and gel electrophoresis, combined with F_{ST} (range 0.0010-0.3659) and HWE (χ^2 test, critical value 3.841) for population genetics analysis
- **Analysis:** Evaluated the potential and challenges of HERV-K solo-LTR as a population genetic and forensic marker, identifying primer sites significantly deviating from HWE
- **Outcome:** Completed the full experimental design, lab work, and data analysis workflow; results received distinction

Bioinformatic Analysis of OCA1A Albinism

London 2023.10 - 2023.11

NCBI, OMIM, Ensembl, UniProt, AlphaFold, ClinVar

- **Structural Analysis:** Analyzed the effect of TYR gene mutations on tyrosinase structure and function, focusing on copper-binding domain mutations, using AlphaFold structural predictions

- **Database Integration:** Performed variant annotation and population-frequency analysis across NCBI, OMIM, Ensembl, UniProt, and ClinVar, covering population-frequency data for 85,874 samples
- **Outcome:** Completed a bioinformatics report, developing skills in multi-database integration, protein structure prediction, and evolutionary conservation analysis

Review: Applications and Limitations of Inbred Mouse Strains in Medical Research London 2023.11 - 2023.12

Literature review, genetic analysis

- **Evidence Synthesis:** Systematically described the genetic principles of inbred mouse strains (>98% allelic homozygosity after 20+ generations) and their use in preclinical drug research, e.g. 40 BALB/c mice evaluating a melittin-loaded nanoliposome formulation against free melittin for breast cancer (renal/liver enzyme assays)
- **Critical Analysis:** Critically analyzed the inbred-mouse model's limitations – genetic/physiological differences from humans, inbreeding depression, and ethical concerns; review received distinction

Skills & Other

- **Molecular & Biochemical Techniques:** PCR and gel electrophoresis (multi-primer genotyping, e.g. 5-primer-set solo-LTR detection), DNA/RNA extraction (Qiagen kits), protein electrophoresis, Western blotting; population genetics analysis (F_{ST} , Hardy-Weinberg equilibrium/ χ^2 tests); variant annotation & protein structure analysis across NCBI, OMIM, Ensembl, UniProt, ClinVar, and AlphaFold
- **Data Analysis:** R (ggplot2, dplyr), Python, statistical inference (GLM, Wilcoxon signed-rank, t-tests, Pearson correlation), quality-control documentation standards
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, skiing